EXECUTION WORKSHEET

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Nested ESXi Host Rebalance

Maintenance-window change sheet — cold resize of the four nested hosts and heavy-VM placement, executed with the stack down.

RIGHT-SIZING | COLD RESIZE | VM PLACEMENT | VSAN | HA / DRS

v1.2 VMware Cloud Foundation 9

Nested ESXi Host Rebalance — Execution Worksheet

Prepared by: Virtual Control LLC **Date:** June 8, 2026 **Document Version:** 1.2 **Classification:** Internal

Use this sheet at the machine. All four nested ESXi hosts are powered off — this is the maintenance window. The work is a **cold resize** of the host VMs in VMware Workstation followed by **manual heavy-VM placement** on startup. It is a *rebalance*, not a capacity increase: total nested RAM stays at **352 GB** (under the 384 GB physical ceiling).

Goal: relieve esxi03 (was ~93% RAM) and put the three heavy VMs — `aria`, `nsx-manager`, `vcenter` — on **three separate hosts**. Companion to the full *Nested ESXi Host CPU/RAM Rebalance Plan (2026-06-08)*.

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1. Pre-Resize — Confirm

- [ ] All four nested ESXi host VMs show **Powered Off** in VMware Workstation.
- [ ] You have the orchestration-host IP from the shutdown wizard (in case vCenter doesn't auto-return on power-on).
- [ ] DRS is **Partially Automated** (it will be re-confirmed once vCenter is back — nothing should auto-move).

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2. The Resize (Workstation, hosts off)

For each VM: **VM** → **Settings** → **Hardware** → **Memory** and → **Processors**.

Host	RAM now → target	vCPU now → target	Change	Done
esxi01	88 → **96 GB**	16 → 16	**+8 GB** (room for NSX Manager)	[ ]

Host	RAM now → target	vCPU now → target	Change	Done
esxi02	64 → **72 GB**	12 → 12	**+8 GB**	[ ]
esxi03	**88 GB** (keep)	**12** (keep)	**no hardware change**	[ ]
esxi04	112 → **96 GB**	32 → **24**	**−16 GB, −8 vCPU** (aria over-provision)	[ ]

Floor: do not size a host carrying a heavy VM below **~64 GB** (ESXi + vSAN overhead + ~20% headroom). If a VM later refuses to power on, raise that host and trim **esxi02** to keep the total at 352.

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3. Arithmetic Check

Confirm **before** powering anything on:

```
 esxi01  96
esxi02  72
esxi03  88
esxi04  96
------------
TOTAL  352 GB   (= today's total; under the 384 GB ceiling — leaves ~32 GB for Windows + Workstation)

vCPU: 16 + 12 + 12 + 24 = 64  (was 72)
```

- [ ] RAM sum reads **352 GB**. If it doesn't, fix it before proceeding.

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4. Power On & Bring vSAN Back

The cluster was shut down with the **vSAN Shutdown Cluster wizard**, so this is the auto-restart path — do **not** boot vCenter by hand.

- [ ] Power on all four nested ESXi host VMs in Workstation.
- [ ] Wait until all four are pingable (`192.168.1.74 / .75 / .76 / .82`).
- [ ] Hosts detect the wizard shutdown state and **auto-restart the cluster + power vCenter on** (give it 5–10 min).
- [ ] If vCenter does not return on its own: open the **orchestration host's** ESXi Host Client (`https://<orchestration-host-ip>/ui`) and trigger **Restart Cluster** from there.
- [ ] Once vCenter is up: vSAN **Skyline Health** clean and **Resyncing Objects = 0** before placing VMs.

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5. Verify Live State, Then Place

The host resize only *supports* the real fix: getting the heavy VMs onto separate hosts. **Do not cold-migrate from the table below until you've confirmed the actual live placement** — then move only what's genuinely off-target.

5.1 Verify live state (read-only — gate)

Once vCenter is reachable:

```
Connect-VIServer vcenter.lab.local -User 'administrator@vsphere.local' -Password '<sso-pw-from-vault>'

# Actual VM -> host placement — THIS is what we act on
Get-VM | Sort VMHost | Select Name, VMHost, PowerState, NumCpu, MemoryGB

# Datastore inventory + free space (for the deferred Aria-to-vSAN question)
Get-Datastore | Select Name, Type, `
  @{n='CapGB';e={[math]::Round($_.CapacityGB)}}, `
  @{n='FreeGB';e={[math]::Round($_.FreeSpaceGB)}} | Sort Name
```

- [ ] Recorded the **actual** current host of every management VM.
- [ ] vSAN Skyline Health clean, **Resyncing Objects = 0**.
- [ ] Compared actual vs. the §5.2 target — **move only the VMs that differ** (don't assume the table; the restart may have placed things differently).

5.2 Target placement

VM	RAM	Target host	Note
nsx-manager	42	**esxi01**	key move — relieves esxi03
vcenter	24	**esxi03**	stays (auto-returns here, ~49%)
aria	54	**esxi04**	stays (only host that powers on 54 GB)
sddc-manager	16	esxi01	with nsx-manager
idm	13	esxi03	with vcenter
vrslcm	6	esxi03	with vcenter
vcf-ops	18	esxi02	—
fleet	5	esxi02	—

Projected at-rest load: esxi01 ~58 GB (60%) · esxi02 ~24 GB (33%) · esxi03 ~43 GB (49%) · esxi04 ~56 GB (58%). No host hot.

5.3 Relocate the off-target VMs

vMotion verified working 2026-06-08 — `idm` and `vrslcm` live-migrated cleanly, pushing past the old ~56% stall point with no downtime (ping never dropped). So **live vMotion is the preferred method now**; cold is only a fallback.

Right-click VM → **Migrate** → **"Change compute resource"** → target host:

- **Live (preferred):** VM stays powered on; vMotion copies memory, then switches over instantly. Zero downtime.
- **Cold (fallback):** if a vMotion ever stalls, power off → **"Change compute resource only"** → power on (storage stays on vSAN, no live migration needed).

Actual movers on the 2026-06-08 bring-up (DRS had scattered them onto esxi04):

- [ ] idm → **esxi03**
- [ ] vrslcm → **esxi03**
- [ ] sddc-manager → **esxi01**
- [ ] vcf-ops → **esxi02**

(nsx-manager already came up on esxi01, vcenter on esxi03, fleet on esxi02 — no move needed.)

5.4 Lock placement with soft should-run rules

DRS is Partially Automated, which **auto-picks the host at power-on**. On the 2026-06-08 bring-up that **scattered the light VMs** (idm, vrslcm, sddc-manager, vcf-ops) onto esxi04 and overcommitted it (112/96 GB) — so rule **every** VM, not just the heavies. Soft rules make placement deterministic and survive future reboots, while still letting **HA restart a VM elsewhere** on a host failure (do **not** use hard "must run" rules — they break HA failover).

Cluster → Configure:

1. **VM/Host Groups → Add → Host Group:** `grp-esxi01` = {esxi01}, `grp-esxi02` = {esxi02}, `grp-esxi03` = {esxi03}, `grp-esxi04` = {esxi04}.
2. **VM/Host Groups → Add → VM Group:** - `vmg-esxi01` = {nsx-manager, sddc-manager} - `vmg-esxi02` = {fleet, vcf-ops} - `vmg-esxi03` = {vcenter, idm, vrslcm} - `vmg-esxi04` = {aria}
3. **VM/Host Rules → Add → "Virtual Machines to Hosts" → "Should run on hosts in group":** - `vmg-esxi01` → `grp-esxi01` - `vmg-esxi02` → `grp-esxi02` - `vmg-esxi03` → `grp-esxi03` - `vmg-esxi04` → `grp-esxi04`

- [ ] 4 host groups + 4 VM groups + 4 **soft** ("should run on") rules created — every VM has a home, so a reboot won't pile them onto esxi04 again.

5.5 Power on in order

`vcenter` (already up) → `sddc-manager` → `nsx-manager` → `vcf-ops` → `fleet` → `vrslcm` → `idm` → **`aria` last**.

- [ ] Each VM landed on its target host (re-run the 5.1 `Get-VM` to confirm).

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6. Post-Change Verification

- [ ] **vSAN:** object health clean, **0 inaccessible**, resync = 0.
- [ ] **Hosts:** each RAM% at rest ~33–60%, none hot; HA all `connectedToMaster` / `master`.
- [ ] **Placement:** `aria` / `nsx-manager` / `vcenter` confirmed on **three separate hosts**.
- [ ] **HA admission control:** reserved % dropped from 34/25 toward ~25/25.
- [ ] **Workloads:** vCenter, NSX, VCF Operations, Aria all back and healthy.
- [ ] **DRS:** left **Partially Automated** (do not return to Fully Automated until vMotion/storage is healthy).

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7. Rollback

Rollback is simple — the only host change is RAM/vCPU numbers in Workstation; **no data is moved or destroyed**.

1. Power off the nested host VMs.
2. Restore the previous values: esxi01 88 / esxi02 64 / esxi03 88 / esxi04 112 GB; esxi04 back to 32 vCPU.
3. Power back up and bring the stack up in order.

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8. Document Information

Field	Value
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Related Documents	Nested ESXi Host CPU/RAM Rebalance Plan (2026-06-08); VCF_Lab_Shutdown_Startup_Guide
Distribution	Lab notebook

Revision History

Version	Date	Notes
1.0	2026-06-08	Initial execution worksheet derived from the rebalance plan: pre-resize confirmation, the four Workstation hardware edits (esxi01 +8, esxi02 +8, esxi03 unchanged, esxi04 −16 GB / −8 vCPU), arithmetic check (352 GB total), vSAN-wizard auto-restart power-on, heavy-VM placement (nsx→esxi01, vcenter stays esxi03, aria stays esxi04), verification, and rollback.
1.1	2026-06-08	Reworked §5 into **Verify Live State, Then Place**: added a read-only live-state gate (`Get-VM` placement + `Get-Datastore` free space) so moves are driven by actual state rather than the 3-day-old notes; added the cold-relocate ("Change compute resource only", no vMotion) steps; added exact **soft should-run VM-Host rule** steps (host groups, VM groups, "should run on hosts in group") for nsx→esxi01 / vcenter→esxi03 / aria→esxi04; and the power-on-order confirmation.
1.2	2026-06-08	Updated §5 to match execution. **vMotion verified working** (idm/vrslcm live-migrated clean past the old ~56% stall) — §5.3 now prefers **live vMotion** (cold as fallback) and lists the actual movers (idm/vrslcm→esxi03, sddc-manager→esxi01, vcf-ops→esxi02; nsx/vcenter/fleet needed no move). **§5.4 broadened to rule every VM grouped by host** (4 host groups + 4 VM groups + 4 soft rules) after the bring-up scattered the *light* VMs onto esxi04 and overcommitted it (112/96 GB) — heavy-only rules wouldn't have prevented that.

Document End

See also: - Nested ESXi Host CPU/RAM Rebalance Plan (2026-06-08) — full rationale, constraints, and decision record - *VCF_Lab_Shutdown_Startup_Guide* — the shutdown/startup order, including the vSAN Shutdown Cluster wizard